

Multiple Choice (10 marks)

1. Suppose we like to calculate $P(H|E, F)$ and we have no conditional independence information. Which of the following sets of numbers are sufficient for the calculation?
 - (a) $P(E, F), P(H), P(E|H), P(F|H)$
 - (b) $P(E, F), P(H), P(E, F|H)$
 - (c) $P(H), P(E|H), P(F|H)$
 - (d) $P(E, F), P(E|H), P(F|H)$
2. Which of the following can only be used when training data are linearly separable?
 - (a) Linear hard-margin SVM.
 - (b) Linear Logistic Regression.
 - (c) Linear Soft margin SVM.
 - (d) The centroid method.
3. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
4. K-fold cross-validation is
 - (a) linear in K
 - (b) quadratic in K
 - (c) cubic in K
 - (d) exponential in K
5. Which of the following sentence is FALSE regarding regression?
 - (a) It relates inputs to outputs.
 - (b) It is used for prediction.
 - (c) It may be used for interpretation.
 - (d) It discovers causal relationships

True / False (10 marks)

Answer True or False

6. True or False: DCGANs primarily rely on self-attention mechanisms to enhance image generation quality and training stability.
7. True or False: The classification run time of the nearest neighbors algorithm is $O(\log N)$, which suggests a logarithmic search efficiency.
8. True or False: The use of weak classifiers in bagging helps prevent overfitting by averaging their predictions.
9. True or False: A Bayesian network with the structure $H \rightarrow U \leftarrow P \leftarrow W$ requires 8 independent parameters to fully specify the joint distribution.
10. True or False: ImageNet contains images of various resolutions, making it a diverse dataset for training deep learning models.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find out the minimum no of swaps required for bracket balancing in the given string.
12. Write a function to select the nth items of a list.

End of Paper